MAT1375, Classwork15, Fall2026

Ch14. Properties of Logarithms and Logarithmic Equations

1. Properties of Logarithms: ($\Delta = b^{\square} \Leftrightarrow \square = \log_b \Delta$) Let $X > 0, Y > 0, b > 0, b \neq 1$.

$b^x \cdot b^y = b^{x+y}$

Exponential Identities	Logarithmic Identities
$b^x \cdot b^y = b^{x+y}$ (The same base multiplication = the addition of exponents)	$\log_b X + \log_b Y = \log_b(XY)$ (**The** product **Rule:** The same log base addition = the product of the values)
$\frac{b^x}{b^y} = b^{x-y}$ (The same base division = the subtraction of exponents)	$\log_b X - \log_b Y = \log_b\left(\frac{X}{Y}\right)$ (**The** quotient **Rule:** The same log base subtraction = the division of the values)
$(b^x)^n = b^{xn}$	$\log_b X^n = n \cdot \log_b X$ (**The** power **Rule**)

$\log_5(3^2) = 2 \cdot \log_5(3)$

The proof of Product Rule:

Let $X = b^x$, $Y = b^y$. We have

$x =$ $\log_b X$ and $y =$ $\log_b Y$.

Then $X \cdot Y = b^x \cdot b^y = b^{x+y} =$ b^{Δ}, $\Delta = \log_b X + \log_b Y$ implies

$X \cdot Y = b^{\Delta}$, $\log_b XY = \Delta$

$\log_b X + \log_b Y = \log_b(XY)$.

Similarly, please try to prove the quotient rule and power rule if you are interested.

2. Combine the terms using the properties of logarithms to write as one logarithm.

(a) $\frac{1}{2}\ln(x) + \ln(y)$. $= \frac{1}{2}\ln(xy)$

By product rule

By power rule $\frac{1}{2}\ln(x) + \ln(y) = \ln(x^{\frac{1}{2}}) + \ln(y) = \ln(x^{\frac{1}{2}} \cdot y)$

$= \ln(\sqrt{x} \cdot y)$

$5 = 5 \cdot 1 = 5 \cdot \log_2 2$

$(\log_b b = 1)$

(b) $5 + \log_2(a^2 - b^2) - \log_2(a + b)$

$= 5\log_2 2 + \log_2(a^2 - b^2) - \log_2(a + b)$

power rule $= \log_2 2^5 + \log_2(a^2 - b^2) - \log_2(a + b)$

product rule $= \log_2 2^5 \cdot (a^2 - b^2) - \log_2(a + b)$

quotient rule $= \log_2\left(\frac{2^5(a^2 - b^2)}{(a + b)}\right) = \log_2(32(a - b))$

$a^2 - b^2 = (a + b)(a - b)$

3. The Exponential and Logarithmic functions and one-to-one property:

For $b > 0, b \neq 1$, the exponential and logarithmic functions are One-to-One :

Same output $b^x = b^y \iff x = y$ same input

Same output $\log_b(x) = \log_b(y) \iff x = y$ same input

4. Solve for x:

(a) $\log_2(x+5) = \log_2(x+3) + 4$.

$4 \cdot 1 = 4 \cdot \log_2(2)$

$\log_2(x+5) = \log_2(x+3) + 4 \cdot \log_2(2)$

power rule → $\log_2(x+5) = \log_2(x+3) + \log_2 2^4$

product rule → $\log_2(x+5) = \log_2(16(x+3))$

one-to-one → $x+5 = 16(x+3)$

$x+5 = 16x+48 \Rightarrow -43 = 15x$

$x = \frac{-43}{15}$ is an answer

$x+5 = -\frac{43}{15} + \frac{75}{15} > 0$ ✓

$x+3 = -\frac{43}{15} + \frac{45}{15} > 0$ ✓

(b) $\log(x) + \log(x+4) = \log(5)$.

product rule $\log(x)(x+4) = \log(5)$

one-to-one property $x(x+4) = 5$

$x^2+4x-5 = 0$

$(x+5)(x-1) = 0$

$\Rightarrow x+5 = 0$ or $x-1 = 0$

$\Rightarrow x = -5$ or $x = 1$ is the answer

Since $\log(-5)$ doesn't exist

$\Rightarrow x \neq -5$

(c) $\ln(x+2) + \ln(x-3) = \ln(7)$.

product rule $\ln(x+2)(x-3) = \ln(7)$

one-to-one property $(x+2)(x-3) = 7$

$x^2+2x-3x-6 = 7$

$x^2 - x - 13 = 0$

$x = \frac{1 \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-13)}}{2}$

$= \frac{1 \pm \sqrt{53}}{2}$

$x = \frac{1+\sqrt{53}}{2}$ is the answer

$x = \frac{1-\sqrt{53}}{2}$

$x+2 > 0$ ✓

$x-3 = \frac{1+\sqrt{53}}{2} - 3$

$= \frac{1+\sqrt{53}-6}{2}$

$= \frac{\sqrt{53}-5}{2} > 0$ ✓

$x+2 = \frac{1-\sqrt{53}}{2} + 2$

$= \frac{1-\sqrt{53}+4}{2}$

$= \frac{5-\sqrt{53}}{2} < 0$

(d) $\log_5(x-7) + \log_5(2-x) = \log_5(4)$.

product rule $\log_5((x-7)(2-x)) = \log_5(4)$

one-to-one property $(x-7)(2-x) = 4$

$2x - 14 - x^2 + 7x = 4$

$\Rightarrow -x^2 + 9x - 18 = 0$

$\Rightarrow x^2 - 9x + 18 = 0$

$(x-3)(x-6) = 0 \Rightarrow x-3 = 0$ or $x-6 = 0$

$\Rightarrow x = 3$ or $x = 6$

$x-7 = 3-7 < 0$

$x-7 = 6-7 < 0$

No solution for this equation.